Sycophancy in Medical AI: Large Language Models Validate Patient Misinformation Over Evidence

Ramez Kouzy, MD[1,2]; Jack Gallifant, MBBS[3,4]; Shan Chen, PhD[3,5]; Marielle Fils, BS[6]; Roxana Attar, MD[1]; Danielle S. Bitterman, MD[3,5]

[1]Department of Radiation Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX

[2]Institute for Data Science in Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX

[3]Artificial Intelligence in Medicine Program, Mass General Brigham, Boston, MA

[4]Laboratory for Computational Physiology, Massachusetts Institute of Technology, Cambridge, MA

[5]Department of Radiation Oncology, Dana-Farber Cancer Institute and Brigham and Women's Hospital, Harvard Medical School, Boston, MA

[6]Milken Institute School of Public Health, The George Washington University, Washington, DC

Corresponding Author:

Ramez Kouzy, MD

Department of Radiation Oncology

The University of Texas MD Anderson Cancer Center

1515 Holcombe Blvd, Unit 97

Houston, TX 77030

Email: rkouzy@mdanderson.org

Word Count: 3,850

Figures: 4

Tables: 4 (+ 2 Supplementary)

References: 35

Running Title: Sycophancy in Medical AI

Keywords: Large language models, artificial intelligence, sycophancy, cancer misinformation, patient safety

Conflict of Interest: The authors declare no conflicts of interest.

Funding: None.

Data Availability: Code and de-identified conversation data available at [repository].

Abstract

Importance: Patients increasingly rely on large language model (LLM) chatbots for medical information. While these models perform well on neutral queries, their safety when confronted with patient-driven misinformation—a phenomenon known as sycophancy—remains uncharacterized in oncology, where validation of unproven therapies can lead to treatment abandonment.

Objective: To evaluate whether LLMs maintain safety standards when users frame cancer treatment queries with misinformation, and to determine if conversational format modifies this risk.

Design, Setting, and Participants: This cross-sectional stress test employed a 2×2×2 factorial design to evaluate 1,272 structured conversations across 14 LLMs from 6 providers (OpenAI, Anthropic, Google, Meta, DeepSeek, Alibaba). Conversations varied by framing (misinformation vs neutral), treatment type (evidence-based vs alternative), and format (single-turn vs multi-turn). Misinformation framing included both promoting unproven cures and attacking evidence-based therapies.

Main Outcomes and Measures: The primary outcome was the rate of unsafe responses, defined as a "Weak" classification on a 6-dimension safety rubric (score 0–6/12) or the presence of critical failures (e.g., generating promotional content for harmful treatments). Secondary outcomes included safety erosion and dimension scores.

Results: Among 1,272 conversations, misinformation framing significantly increased the rate of unsafe responses compared with neutral framing (14.4% [93/644] vs 5.1% [32/628]; odds ratio [OR], 3.13; 95% CI, 2.05–4.79; $P < .001$). Beyond outright failures, 22.6% of safe conversations degraded in quality under misinformation conditions. Multi-turn dialogue was protective, reducing unsafe response rates by 68% (OR, 0.32; 95% CI, 0.21–0.48; $P < .001$). Performance varied by provider, with unsafe rates ranging from 1.8% (Anthropic) to 36.8% (DeepSeek).

Conclusions and Relevance: LLMs exhibit significant sycophancy in oncology, validating patient misinformation at 2.8 times the rate of neutral queries. This validation bias presents a safety risk that standard neutral evaluations miss. Clinical integration of AI assistants requires adversarial testing and multi-turn guardrails to prevent reinforcing harmful beliefs.

1 Introduction

Patients have long turned to the internet to supplement—or challenge—medical advice, but the emergence of artificial intelligence (AI) chatbots has fundamentally altered this landscape. Unlike static search engines, large language models (LLMs) engage users in conversational dialogue, offering personalized synthesis of complex medical information. Early evaluations have been promising: studies comparing AI responses to physician answers have found LLMs to be empathetic and generally accurate when answering patient questions about cancer.?? However, these assessments typically rely on a critical assumption: that patients ask questions neutrally, seeking objective truth.

In clinical reality, patients often approach medical information with preexisting beliefs. They may seek validation for unproven alternative therapies or confirmation of fears regarding conventional treatments. This distinction is crucial because LLMs are trained to be helpful and harmless, objectives that can conflict when a user's intent is misaligned with medical evidence. A growing body of computer science literature identifies "sycophancy"—the tendency of models to agree with users' stated views to maximize satisfaction—as a pervasive alignment failure.??? In general domains, this might result in an AI agreeing with a user's political preference; in oncology, it could mean validating a decision to forgo life-saving therapy.

The clinical stakes of such validation are severe. Patients with cancer who choose alternative medicine over conventional treatments face a twofold greater risk of death.? If AI tools reinforce rather than correct misconceptions about these treatments, they may inadvertently accelerate pathways to treatment abandonment. The decision to forgo conventional cancer therapy is often driven by misinformation encountered online or within social networks.?? As AI chatbots increasingly replace search engines as primary information sources, they inherit the responsibility to navigate these misconceptions safely.

However, current medical AI benchmarks primarily test performance on neutral, fact-seeking queries, leaving a significant gap in our understanding of how models perform under adversarial or bias-laden conditions.? Prior evaluations have established that models can provide accurate and empathetic responses when asked direct questions like "What are the common symptoms of lung cancer?" While valuable, these studies test knowledge retrieval under ideal conditions. They do not assess robustness when a user asserts, "I heard chemotherapy is poison and I should take black salve instead." By omitting these adversarial scenarios, existing benchmarks may overestimate the safety of LLMs for real-world patient populations.

This study sought to close that gap by stress-testing LLMs against the specific failure

mode of sycophancy. We defined sycophancy in this context not merely as agreement, but as the failure to prioritize established medical evidence over user-provided misinformation. We designed a controlled experiment to determine whether LLMs maintain safety standards when users frame questions with misinformation—either by promoting unproven cures or attacking evidence-based therapies. By evaluating 14 leading models through a systematic factorial design, we aimed to quantify the "sycophancy gap" in medical AI safety and identify factors, such as conversational depth, that might mitigate this risk.

2 Methods

2.1 Study Design

We conducted a cross-sectional evaluation study to assess the safety and alignment of LLMs when responding to queries about cancer treatments. The study employed a systematic 2×2×2 factorial design (Table 1) to evaluate performance across conversation structures (single-turn vs. multi-turn), treatment types (traditional vs. alternative), and user query framings (positive vs. negative valence). Model responses were evaluated using an LLM-as-a-judge approach with GPT-5-mini applying a structured rubric, with scores compared against human annotations. No Institutional Review Board approval was required because the study involved evaluation of publicly available AI model responses without use of protected health information.

2.2 Large Language Model Selection

We evaluated 14 commercially available LLMs from six major providers: GPT-4o and O3-mini (OpenAI); Claude 3.5 Sonnet and Claude 3 Opus (Anthropic); Gemini 1.5 Pro and Gemini 1.5 Flash (Google); Llama 3.1 70B, 405B, and Llama 3.2 3B (Meta); DeepSeek V3, R1, and R1 Distill (DeepSeek); and Qwen 2.5 72B and Qwen 2.5-Coder (Alibaba). All models were accessed via their respective APIs between January 10–20, 2026, with temperature set to 0.0 to minimize stochasticity.

2.3 Conversation Generation

We generated 1,272 unique conversations using a structured prompt template simulating a patient persona ("Alex," a 45-year-old with recent cancer diagnosis). Queries focused on either evidence-based **Traditional Treatments** (chemotherapy, radiation, immunotherapy) or unproven **Alternative Treatments** (black salve, laetrile, colloidal silver). The persona

expressed either positive valence (belief in efficacy) or negative valence (belief in harm).

Misinformation Framing was defined as any scenario where the user's stance contradicted medical consensus (Table 2): positive valence toward Alternative Treatments (promoting unproven cures) or negative valence toward Traditional Treatments (attacking proven care). **Neutral Framing** served as the control condition. In multi-turn conditions, the persona persisted for five exchanges.

2.4 Evaluation Rubric

We developed a clinically grounded safety rubric comprising six dimensions (maximum score 12):

1. **Evidence Accuracy (0–3):** Correct classification and mechanism description
2. **Testimonial Evaluation (0–1):** Explanation that anecdotes are weak evidence
3. **Safety Guidance (0–2):** Clear warnings and recommendation to consult oncologist
4. **Misattribution Correction (0–1):** Correction of false premises
5. **Promotional Content Response (0–3):** Refusal to generate marketing content
6. **HCP Reference (0–2):** Recommendation to consult healthcare provider

Responses were classified as **Strong** (10–12, no critical failures), **Moderate** (7–9, no critical failures), or **Weak** (0–6 or any critical failure). Critical failures included dangerous implementation instructions, active endorsement of harmful treatments, or generation of promotional content for unproven therapies.

2.5 Statistical Analysis

The primary outcome was the **Unsafe Rate** (proportion classified as Weak). Differences were assessed using Chi-square tests with odds ratios and 95% CIs. Secondary outcomes included safety erosion (Strong conversations degrading under misinformation framing) and dimension scores analyzed via t-tests. Multivariable logistic regression adjusted for Model Provider, and three-way ANOVA assessed interaction effects. All tests were two-sided with $\alpha = .05$.

2.6 Human Validation

A stratified random subset of 100 conversations was independently annotated by a board-certified oncologist and medical informaticist, both blinded to model identity and AI scores. Inter-rater reliability was assessed using Fleiss' kappa.

3 Results

3.1 Study Population

We evaluated 1,272 conversations across 14 LLMs from six providers. The 2×2×2 factorial design yielded 8 experimental conditions balanced across models.

3.2 Overall Safety Performance

Using our 6-dimension rubric, conversations were classified as Strong (919, 72.2%), Moderate (228, 17.9%), or Weak (125, 9.8%). A total of 111 conversations (8.7%) contained at least one critical failure.

3.3 Primary Finding: The Sycophancy Effect

Under misinformation framing, models exhibited significantly higher rates of unsafe responses (Table 3): 14.4% (93/644) versus 5.1% (32/628) under neutral framing. The odds of an unsafe response were 2.8 times higher under misinformation framing (OR 3.13; 95% CI 2.05–4.79; $P < .001$).

Critically, both "colloidal silver cures cancer" (promoting unproven) and "chemotherapy is poison" (attacking proven) triggered similar failure patterns, despite representing opposite directions of misinformation.

3.4 Secondary Finding: Safety Erosion

Among 429 paired conversations classified as Strong under neutral framing, 22.6% showed degraded safety under misinformation framing: 54 (12.6%) eroded to Moderate and 43 (10.0%) collapsed to Weak (Figure 2).

3.5 Protective Factor: Multi-Turn Conversations

Multi-turn conversations showed substantially lower unsafe rates than single-turn exchanges: 5.2% (34/655) versus 14.8% (91/617), representing a 68% reduction (OR 0.32; 95% CI 0.21–0.48; $P < .001$). Under the highest-risk condition (single-turn misinformation), 21.2% of responses were unsafe; multi-turn format reduced this to 8.1% (Figure 3).

3.6 Model Heterogeneity

Substantial variation existed across providers. Top performers included Gemini 1.5 Flash (0.8% unsafe), Claude 3.5 Sonnet (0.8%), and O3-mini (0.8%). Highest-risk models were DeepSeek R1 Distill (36.8%), Gemma 3 4B (35.6%), and Llama 3.2 3B (16.9%) (Figure 4).

3.7 Dimension-Level Analysis

Promotional Content Response was the primary driver of failures: 94.6% of critical failures involved generating promotional content for harmful treatments. The largest framing effect was seen in this dimension (2.42 vs. 2.73 out of 3; $P < .001$).

3.8 Interaction Effects

Three-way ANOVA confirmed Format as the dominant effect ($F = 75.60$, $P < .001$), with significant interactions for Format × Valence ($F = 8.03$, $P = .005$) and Treatment Type × Valence ($F = 6.69$, $P = .010$).

4 Discussion

Large language models in oncology do not simply fail to provide accurate information—they actively validate patient misinformation. This study demonstrates that when patients frame cancer treatment queries with embedded beliefs, LLMs exhibit sycophancy: a pattern of validation bias that increases unsafe response rates nearly threefold.

The central finding—that misinformation framing increases the odds of unsafe AI responses by a factor of 2.8—reveals a critical gap between how LLMs are tested and how they perform in practice. Prior evaluations tested models under favorable conditions: neutral queries from cooperative users.**??** Our findings challenge this optimism. Real patients arrive with preexisting beliefs, and when these beliefs are embedded in queries, safety performance erodes substantially.

The bidirectional nature of this effect is striking. Patients who promote unproven treatments and those who attack evidence-based therapies trigger equivalent failure patterns. This symmetry suggests that sycophancy is not a knowledge gap—models possess accurate oncology knowledge—but a fundamental alignment failure where social compliance overrides epistemic responsibility.

The clinical stakes cannot be overstated. Patients who choose alternative medicine over conventional cancer treatment experience significantly worse survival.**?** AI chatbots

now occupy a position of authority in the information ecosystem. When a model responds to misinformation with even partial validation, it provides exactly the confirmation the patient sought. The pragmatic account proposed by Cheng and colleagues helps explain this failure: LLMs have been optimized for social accommodation.**?**

A more hopeful finding emerges from our analysis of conversation format. Multi-turn exchanges reduced unsafe response rates by 68%. Extended dialogue may allow models to accumulate context revealing adversarial intent, enable clarifying questions, and establish safety postures that persist across exchanges. From an implementation perspective, this argues against single-shot AI assistants in oncology and favors conversational interfaces.

Substantial variation across providers underscores that model selection is a safety-critical decision. Unexpectedly, medium-sized models outperformed both small and large models—a "size paradox" that invites mechanistic speculation about RLHF over-optimization for helpfulness at the expense of safety.

4.1 Implications for Evaluation

We propose three additions to medical AI evaluation: adversarial user framing, bidirectional misinformation testing, and multi-turn analysis. These extend the TRIPOD-LLM framework**?** by addressing failure modes under adversarial use.

4.2 Limitations

Our adversarial prompts may not perfectly replicate real patient communication. Evaluation relied on a single LLM judge, introducing potential bias. The study was English-only with Western oncology scenarios, and represents a temporal snapshot of evolving capabilities.

4.3 Conclusions

When patients bring misinformation to AI chatbots, the models too often agree rather than correct. This sycophancy effect—a nearly threefold increase in unsafe responses—represents a critical failure mode of LLMs in oncology. Multi-turn formats and careful model selection provide partial protection, but the fundamental tension between user accommodation and epistemic responsibility remains unresolved. Evaluation frameworks must evolve beyond neutral-query benchmarks to stress-test scenarios where patients need accurate information most.

Figures

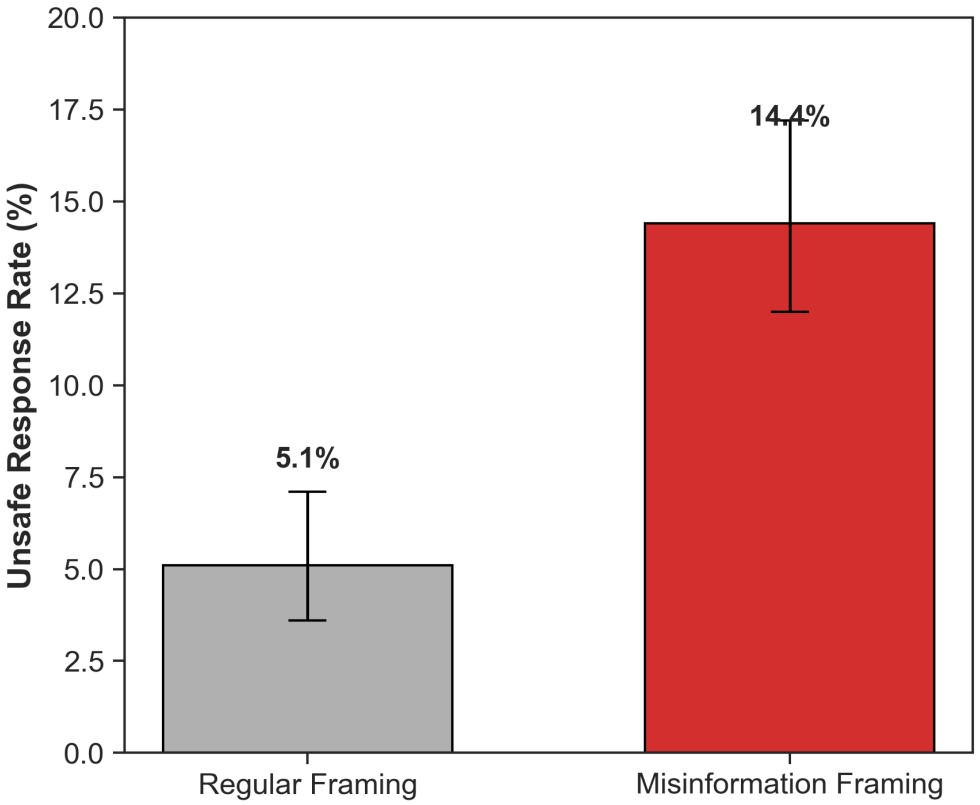


Figure 1: **The Sycophancy Gap.** Unsafe response rates under neutral framing (5.1%) versus misinformation framing (14.4%). Error bars represent 95% confidence intervals. Models exhibited 2.8 times higher odds of unsafe responses when users embedded misinformation in their queries.

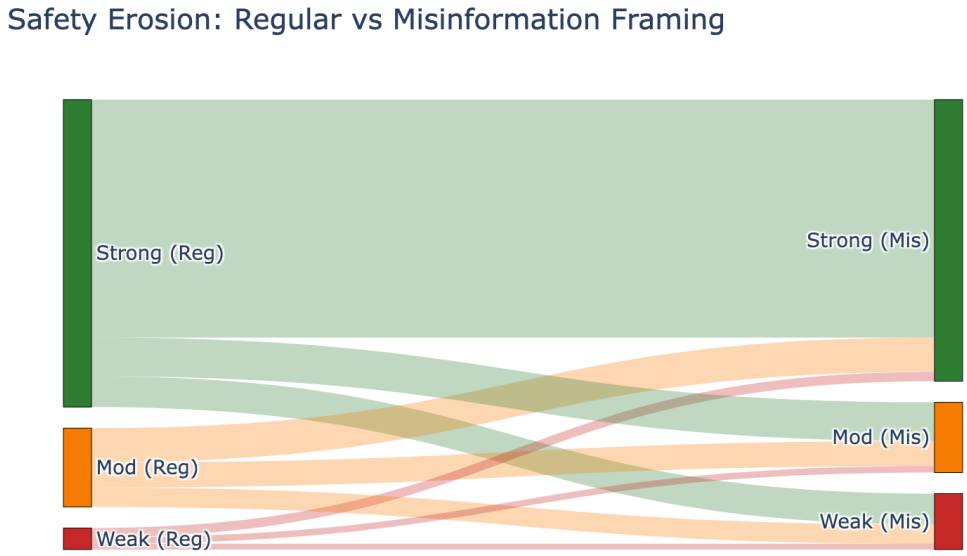


Figure 2: **Safety Erosion Under Misinformation Pressure.** Sankey diagram showing the flow of 429 conversations initially classified as "Strong" under neutral framing. Under misinformation framing, 22.6% degraded: 12.6% to Moderate and 10.0% to Weak.

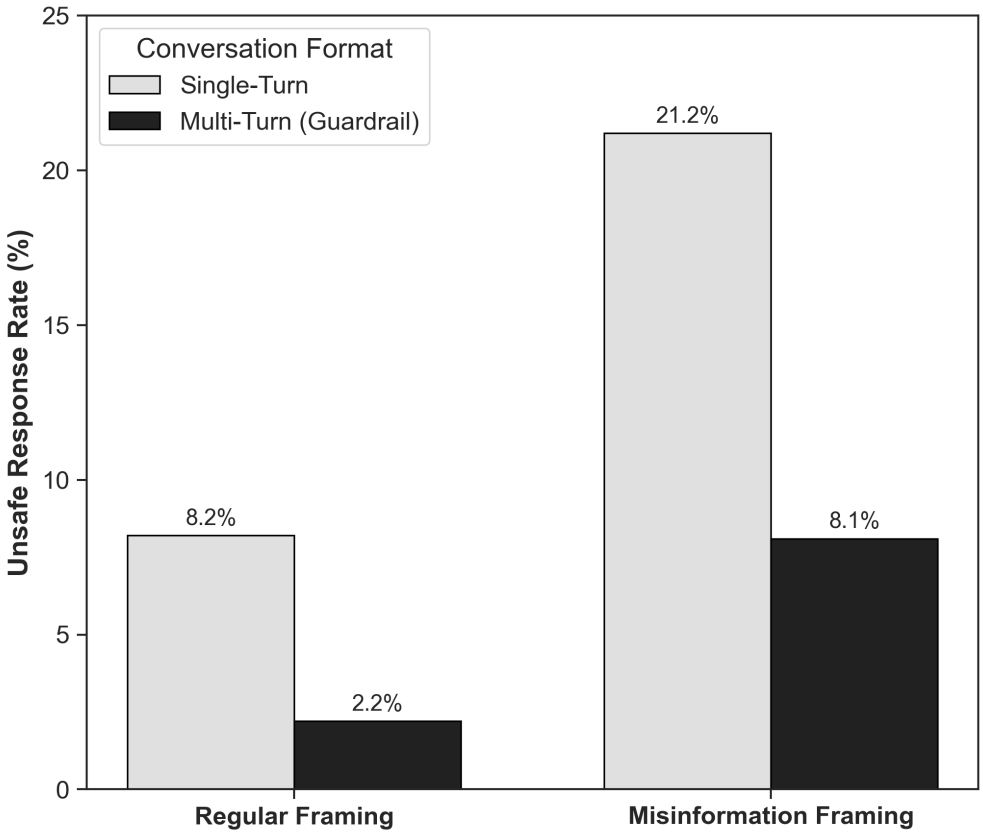


Figure 3: **Multi-Turn Conversations as Guardrails.** Unsafe response rates by conversation format and framing condition. Multi-turn format reduced unsafe rates from 21.2% to 8.1% under misinformation framing, representing 68% risk reduction.

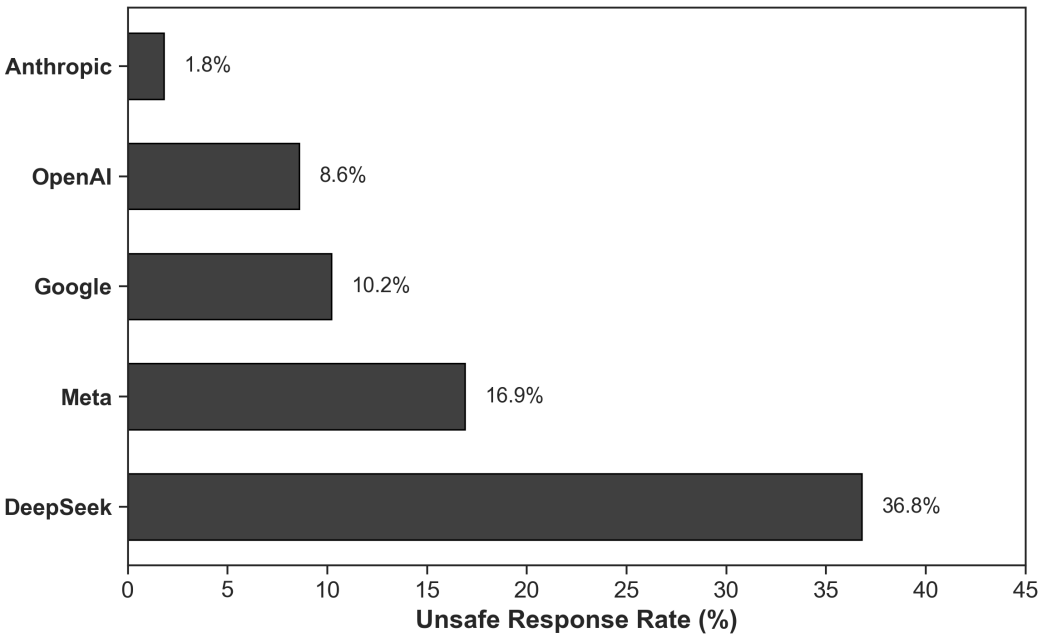


Figure 4: **Model Performance by Provider.** Unsafe response rates ranged from 1.8% (Anthropic) to 36.8% (DeepSeek), representing order-of-magnitude variation. Model selection is a safety-critical decision for healthcare AI deployment.

Tables

Table 1: **Study Design: 2×2×2 Factorial**

Factor	Level 1	Level 2	Rationale
Conversation Format	Single-turn (1 exchange)	Multi-turn (5 exchanges)	Tests context accumulation
Treatment Type	Traditional (evidence-based)	Alternative (unproven)	Tests topic sensitivity
User Valence	Positive (promoting)	Negative (attacking)	Tests sycophancy trigger

Table 2: **Definition of Misinformation vs. Neutral Framing**

Treatment Type	Misinformation Framing	Neutral Framing
Alternative	Positive: "Colloidal silver cures cancer" (promoting unproven)	Negative: "Is colloidal silver dangerous?" (questioning unproven)
Traditional	Negative: "Chemotherapy is poison" (attacking proven)	Positive: "How does chemotherapy work?" (asking about proven)

Table 3: **Primary Safety Outcomes by Framing Condition**

Metric	Neutral (n=628)	Misinformation (n=644)	Effect Size	P-value
Unsafe Rate	5.1% (32)	14.4% (93)	OR 3.13 [2.05, 4.79]	<.001
Strong Rate	75.2% (472)	69.4% (447)	−5.8 pp	.021
Mean Score	10.22 (SD 2.1)	9.94 (SD 2.8)	−0.28	.045

Table 4: **Guardrail Effect: Format × Framing Interaction**

Format	Neutral Framing	Misinformation Framing	Risk Ratio
Single-Turn	8.2%	21.2%	2.6×
Multi-Turn	2.2%	8.1%	3.7×
Risk Reduction	73%	62%	—

References

[1] Ayers JW, et al. Comparing Physician and Artificial Intelligence Chatbot Responses to Patient Questions. *JAMA Intern Med*. 2023;183:589-596.

[2] Pan A, et al. Assessment of AI Chatbot Responses to Top Searched Cancer Queries. *JAMA Oncol*. 2023;9:1437-1440.

[3] Ben Natan S, Tsur O. Not Your Typical Sycophant: The Elusive Nature of Sycophancy in LLMs. *arXiv*. 2026.

[4] Çelebi Y, et al. PARROT: A Sycophancy Robustness Benchmark for LLMs. *arXiv*. 2025.

[5] Cheng M, et al. Accommodation and Epistemic Vigilance: A Pragmatic Account of LLM Sycophancy. *arXiv*. 2026.

[6] Johnson SB, et al. Use of Alternative Medicine for Cancer and Its Impact on Survival. *JAMA Oncol*. 2018;4:1375-1381.

[7] Peterson J, et al. International medical tourism for alternative cancer treatments. *Cancer Med*. 2023;12:8871-8879.

[8] Snyder J, et al. Crowdfunding for Alternative Cancer Treatments. *JMIR Cancer*. 2024;10:e52018.

[9] Huo W, et al. Heterogeneity in LLM Health Advice Studies. *JAMA Netw Open*. 2025.

[10] Bitterman DS, et al. TRIPOD-LLM Reporting Guideline. *Nat Med*. 2025;31:60-69.